Frequency-Guided Denoising Network for Semantic Segmentation of Remote Sensing Images

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***Abstract*—Semantic segmentation of high-resolution remote sensing images (RSIs) remains challenging due to the degradation of high-frequency (HF) semantic cues during convolutional encoding and the lack of frequency consistency in multistage feature fusion. To address these issues, we propose a frequency-guided denoising network (FreDNet) that explicitly enhances frequency-sensitive representations throughout the segmentation process. Specifically, we introduce the denoising-bypassed residual block (DRB), which incorporates a frequency-guided cross-stage fusion module (FDM) and a frequency-aware fusion module (FFM) to suppress frequency-domain noise while preserving edge structures. Furthermore, we design a frequency-guided cross-stage fusion module (FCFM) that leverages frequency intensity response maps to adaptively fuse encoder and decoder features. These components work collaboratively to enhance the frequency robustness and spatial consistency of the segmentation predictions. Extensive experiments on three challenging benchmarks, ISPRS Vaihingen, ISPRS Potsdam, and LoveDA, demonstrate that FreDNet achieves superior performance, surpassing the latest state-of-the-art (SOTA) approaches by up to 0.8% in mean intersection over union (mIoU) and 0.9% in overall accuracy (OA), while maintaining a lightweight inference cost. In addition, ablation study confirms the contribution of each component of FreDNet.**

***Index Terms*—Attention mechanism, convolutional neural network (CNN), frequency domain, remote sensing images (RSIs), semantic segmentation.**

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I. Introduction

SEMANTIC segmentation of remote sensing images (RSIs) refers to the task of assigning a semantic label to each pixel within an image [1], [2], [3], [4], [5]. This task plays a fundamental role in a variety of real-world applications, including urban planning, land-use mapping, agricultural monitoring, environmental management, and disaster response [6], [7], [8], [9], [10], [11], [12]. Compared to natural images, RSIs are generally characterized by complex compositions with high-resolution textures, irregular object shapes, and significant intraclass variation [13], [14], [15], [16]. These properties pose unique challenges for segmentation models, necessitating robust feature learning mechanisms that can capture fine-grained details while preserving structural consistency.

Convolutional neural networks (CNNs) have long been the dominant backbone for semantic segmentation due to their capacity for hierarchical representation learning [17]. Pioneering works such as FCN [18] and SegNet [19] introduced end-to-end trainable architectures for dense prediction. Later improvements, including U-Net [20], DeepLab series [21], and PSPNet [22], enhanced context modeling through dilated convolutions and pyramid pooling. To adapt these designs to remote sensing imagery, architectures like D-LinkNet [23] and DFCN [24] employed encoder–decoder frameworks to maintain spatial precision while capturing multiscale semantics. More recent CNN-based networks such as ResUNet-a [25] and SDNF [26] further tailored convolutional designs for high-resolution overhead scenes. Nevertheless, due to the inherently local nature of convolutional operations, effectively modeling long-range dependencies and preserving high-frequency (HF) structures such as object boundaries and fine contours remain challenging in complex RSI scenarios [27].

To alleviate these limitations, several studies have explored explicit multiscale or multifield-of-view modeling strategies. Representative methods, such as MFVNet [28], introduce adaptive fusion across multiple receptive fields to balance local detail representation and broader contextual information in RSI segmentation. Such designs have demonstrated notable effectiveness in enhancing spatial context modeling, particularly for scenes with significant scale variation. At the same time, their performance and efficiency are often influenced by architectural complexity and input resolution, which motivates continued exploration of alternative or complementary mechanisms for capturing fine-grained structures under practical computational constraints.

To address the locality constraints of convolution, attention mechanisms have been widely adopted in both natural and

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remote sensing segmentation models. Squeeze-and-excitation network (SE-Net) [29] and convolutional block attention module (CBAM) [30] introduced channel and spatial attentions to recalibrate feature maps adaptively. Advanced schemes like DANet [31], OCRNet [32], and NLNet [33] leveraged pixel-level or object-contextual attention to enhance global semantic awareness. In the field of RSI segmentation, attention modules have been integrated into networks to improve semantic alignment and suppress irrelevant background noise [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46]. Despite these advancements, most existing attention-based methods are primarily formulated in the spatial domain and only implicitly interact with frequency-related cues, which are closely associated with boundaries, textures, and fine structural variations in high-resolution RSIs.

With the advent of vision transformers (ViTs), attention modeling has evolved further. ViT [47] introduced a pure transformer-based framework for image classification by modeling long-range dependencies through self-attention. Segmentation-specific variants such as Segmenter [48], SegFormer [49], and Swin Transformer [50] have demonstrated state-of-the-art (SOTA) performance on natural image segmentation benchmarks. These approaches have also inspired adaptations in the RSI domain. Methods like MBTUNet [51], FarSeg++ [52], LETFormer [53], CLCFormer [54], AAFormer [55], and Swin-UNet [56] introduced hybrid designs combining local–global attention and spatial priors for remote sensing segmentation. More recently, holistic learning paradigms exemplified by REST [57] further explore end-to-end processing of large-scale or whole-scene RSIs, providing valuable insights into global semantic consistency and system-level optimization. While such approaches significantly advance scene-level understanding, effectively preserving fine structural details and HF semantics across diverse spatial scales remains an open research topic. In parallel, the rapid development of large-scale remote sensing datasets, such as MEET [58], has underscored the increasing diversity and complexity of real-world RSI distributions. This trend highlights the need for segmentation models that can flexibly balance local structural fidelity and large-scale contextual variation, while maintaining practical efficiency and scalability.

More fundamentally, both CNN-based and transformer-based methods commonly neglect frequency-domain characteristics, despite their critical importance in the semantic segmentation of RSIs. HF components, such as edges, textural patterns, and spatial discontinuities, are often suppressed during downsampling or misclassified as noise during feature transformation. This undermines the accurate reconstruction of fine contours and small-scale object semantics, which are pivotal for detailed segmentation.

Recent attempts to incorporate frequency-domain cues into deep networks have shown promising potential. Methods such as FcaNet [59], FsaNet [60], XNet [61], FreqFusion [62], and FreqMix [63] leverage frequency-aware operations to enrich feature representations. Among them, wavelet-based transforms, such as the 2-D discrete wavelet transform (DWT), offer natural decomposition into multiresolution frequency bands, making them highly suitable for structured spectral modeling. Despite this, integrating frequency priors into deep networks for segmentation remains a nontrivial challenge due to two major issues.

1) *Lack of Frequency-Aware Semantic Preservation During Representation Learning:* In high-resolution RSIs, critical semantic cues such as object contours, texture boundaries, and local discontinuities often reside in HF bands. However, standard convolutional and pooling operations tend to emphasize low-frequency components due to their inherent spatial smoothing effects, which leads to suppression or distortion of HF semantics during feature extraction and transformation. Most existing CNN- and transformer-based segmentation networks are optimized primarily in the spatial domain and lack explicit frequency-aware mechanisms to enhance or retain these structurally informative signals.
2) *Absence of Frequency-Guided Feature Alignment Across Encoder–Decoder Stages:* Encoder–decoder architectures commonly adopt skip connections for feature fusion, yet such strategies, typically involving direct concatenation or elementwise addition, fail to consider spectral compatibility between features at different stages. Without leveraging frequency-domain consistency, these fusion operations may introduce semantic discontinuities or misaligned boundaries, which hinder the recovery of fine-grained details and reduce segmentation precision in complex scenes.

To address these limitations, we propose a novel frequency-guided denoising network, termed FreDNet, for semantic segmentation of RSIs. FreDNet integrates spectral priors into the learning process through DWT-based decomposition and frequency-conditioned modulation. Specifically, we design frequency-aware modules and attention-enhanced blocks to facilitate multistage denoising and detail preservation. Unlike conventional networks that treat HF signals as noise, our approach treats them as semantic cues and learns to selectively enhance or suppress them via structured frequency modulation. The main contributions of this work are summarized as follows.

1) We design a frequency-aware denoising module (FDM) and a frequency-aware fusion module (FFM) that jointly extract and manipulate spectral energy using 2-D DWT. These modules enable frequency-conditioned representation learning by introducing learnable modulation parameters for content-aware feature enhancement and noise suppression.
2) We propose a denoising-bypassed residual block (DRB) to decouple representation learning into two branches, a conventional residual path and a frequency-guided denoising path. The latter integrates FDM and FFM to selectively retain HF semantics. Their fusion helps recalibrate distorted features while preserving structural details.
3) To facilitate cross-stage semantic consistency, we introduce the frequency-guided cross-stage fusion module (FCFM). This module generates spatially adaptive attention maps guided by wavelet-derived intensity

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distributions. These maps act as frequency-aware gates to improve alignment between encoder and decoder features, especially for boundary-sensitive and fine-grained semantics.

4) Extensive experiments on three benchmark datasets, i.e., ISPRS Vaihingen [64], ISPRS Potsdam [65], and LoveDA [66], demonstrate that FreDNet achieves superior segmentation performance compared to several SOTA methods. Ablation studies further validate the effectiveness of each proposed component.

The remainder of this article is organized as follows. Section II reviews related work in semantic segmentation and frequency-aware learning. Section III details the proposed FreDNet architecture and its constituent modules. Section IV presents the experimental results and comparative analysis. Finally, Section V concludes this article and discusses potential directions for future work.

II. Related Works

A. Semantic Segmentation of RSIs

Semantic segmentation of RSIs poses unique challenges compared to natural images due to complex object boundaries, irregular shapes, and high spectral variability among similar classes [67], [68], [69]. These characteristics often lead to ambiguous predictions in densely packed or spectrally overlapping regions, necessitating more robust and structurally aware modeling approaches.

CNNs have served as a cornerstone for RSI segmentation, due to their ability to extract hierarchically structured features from high-resolution imagery [70], [71]. Encoder–decoder frameworks, such as U-Net and its variants, have demonstrated strong capabilities in dense labeling tasks. ResUNet-a [25] introduces residual and attention mechanisms into the U-Net backbone, enabling more effective multiscale fusion and better gradient propagation. To capture fine-grained spatial structures, LANet [37] integrates local attention for patchwise enhancement, while Li et al. [38] design a dual-branch attention pathway that employs spatial attention in shallow layers and channelwise recalibration in deeper layers. Furthermore, hybrid designs such as SCAttNet [39] and HCANet [72] jointly leverage spatial and channel cues to improve saliency discrimination. Dual-path networks including WiCoNet [73] and SAM [74] introduce complementary local–global encoding pipelines to better handle cluttered or heterogeneous environments. While these CNN-based methods have achieved steady progress, their reliance on spatially local convolutional operations and attention mechanisms may pose challenges in consistently modeling long-range dependencies and preserving fine boundary continuity, particularly under resolution degradation or large-scale variations.

In recent years, transformer-based models have gained traction in RSI segmentation due to their self-attention mechanisms that facilitate global dependence modeling [75]. Inspired by ViT [76], several works have extended transformer designs for overhead imagery. For instance, FarSeg++ [52] and LETFormer [53] embed multiscale attention within hierarchical backbones to address scale-aware contextual encoding. Convolution-integrated variants like CLCFormer [54] and AAFormer [55] enhance spatial biasing and adaptive fusion capabilities. Swin-UNet [56] utilizes shifted window-based attention to strike a balance between computational efficiency and contextual expressiveness. Beyond structural innovations, recent efforts explore novel attention pathways: Xiao et al. [77] propose interstage multiscale refinement; CMLFormer [78] incorporates convolutional-local transformers for edge preservation; and Chen et al. [79] introduce hybrid attention fusion in the decoder for contextual reinforcement. Lightweight and knowledge-distilled designs such as [80] target real-time and resource-constrained deployments. Advanced models like hypergraph BiFormer [81] and UM2Former [82] further explore relational modeling and multimodal fusion for hyperspectral scenes. RS3Mamba [3] introduced the Mamba model for better contextual understanding of RSIs. Overall, transformer-based approaches substantially improve global context modeling in RSI segmentation; however, most existing designs primarily operate in the spatial domain, and the interaction between global attention mechanisms and frequency-sensitive structural cues remains relatively underexplored. In addition, patch-based representations and positional encoding strategies may influence spatial coherence when handling high-resolution scenes with complex boundaries.

Beyond architectural design, recent studies have also emphasized the importance of data scale, semantic structure, and higher level reasoning for advancing remote sensing understanding. For example, the STAR benchmark [83] introduces large-size satellite imagery with scene graph annotations, enabling structured scene representation and relational reasoning at scale. In a related direction, Li et al. [84] explore multigranularity knowledge graph reasoning for zero-shot urban land-use mapping, highlighting the role of explicit semantic reasoning when pixel-level supervision is limited. These works extend the scope of remote sensing analysis beyond per-pixel classification, while still relying on accurate and structure-preserving segmentation as a foundational component.

Taken together, although CNN- and transformer-based paradigms have independently advanced RSI segmentation from complementary perspectives, effectively preserving fine-grained structural details while maintaining semantic consistency across scales remains an active area of research. This observation motivates continued exploration of alternative representational dimensions that can complement spatial-domain modeling.

B. Frequency-Domain-Incorporated Semantic Segmentation

Frequency-domain representations have emerged as a compelling complement to spatial features in semantic segmentation, offering enhanced modeling of textures, boundary structures, and HF variations often attenuated by conventional spatial convolutions. By transforming feature maps via Fourier or wavelet transforms, recent approaches aim to encode spectral semantics and reduce sensitivity to low-level distortions or noise.

In the domain of natural image segmentation, several methods have demonstrated the benefits of frequency-domain

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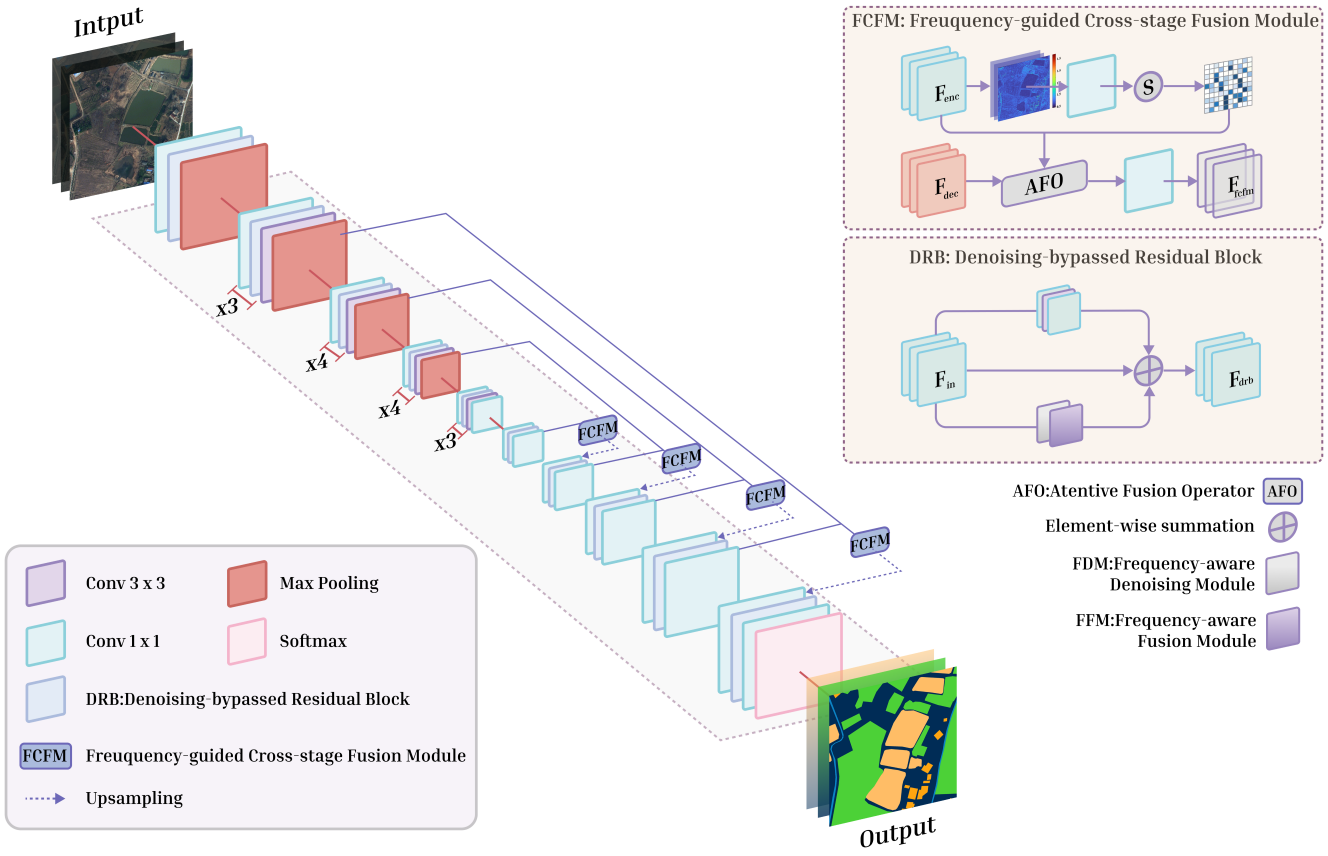

Fig. 1. Overview of the proposed FreDNet.

processing. FsaNet [60] proposes a frequency self-attention mechanism that selectively amplifies informative spectral components while attenuating redundant features. Chen et al. [85] introduce frequency-adaptive dilated convolutions to dynamically adjust receptive fields based on dominant frequencies, improving detail preservation and boundary localization. Follow-up studies [86] further investigate aliasing artifacts in CNNs, while FDA [87] leverages Fourier-based domain adaptation to improve generalization in cross-domain segmentation. Other works such as [88], [89], and [90] explore frequency decoupling for generalization, attention refinement, and interactive learning. FDLNet [91] accelerates segmentation via frequency-domain convolutions, while Rao et al. [92] integrate frequency priors for robust few-shot segmentation.

In parallel, frequency-domain modeling is gaining prominence in the semantic segmentation of RSIs, where the preservation of fine-grained structures is particularly critical. Wavelet-based fusion designs such as SFFNet [93] and FDNet [4] decouple high- and low-frequency components, enabling more effective multiscale feature learning. FSDENet [94] and SSCBNet [95] integrate frequency-guided detail enhancement and spatial priors to recover thin structures and texture-rich regions. Zhang et al. [96] propose a frequency-aware Swin transformer, while AFE-Net [97] leverages adaptive modulation of frequency responses to improve structural accuracy. Moreover, approaches such as SFCANet [98] and [99] explore domain generalization through spectral alignment and frequency-informed attention. Shan et al. [100] further decouple semantic content across frequency bands to improve spatial-contextual balance.

Despite the growing body of work, many existing frequency-domain approaches incorporate spectral cues as auxiliary components or localized enhancements, rather than embedding them as integral elements of the overall network architecture. In addition, frequency-aware representations are often processed independently at individual stages, and their alignment across encoder–decoder pathways is not always explicitly modeled.

Recently, large-scale and foundation-model paradigms have also begun to incorporate frequency-related or multimodal representations to support a broad range of Earth observation tasks. For instance, Wu et al. [101] propose a semantic-enhanced multimodal remote sensing foundation model, demonstrating the potential of unified representations learned from diverse sensor modalities and large-scale data. While such models aim to support multiple downstream tasks, frequency-guided pixel-level segmentation remains a complementary and important direction for enhancing structural fidelity and boundary precision.

Motivated by these observations, this work focuses on integrating frequency-domain cues directly into the core representation learning and fusion processes of semantic segmentation networks, rather than treating them as peripheral enhancements.

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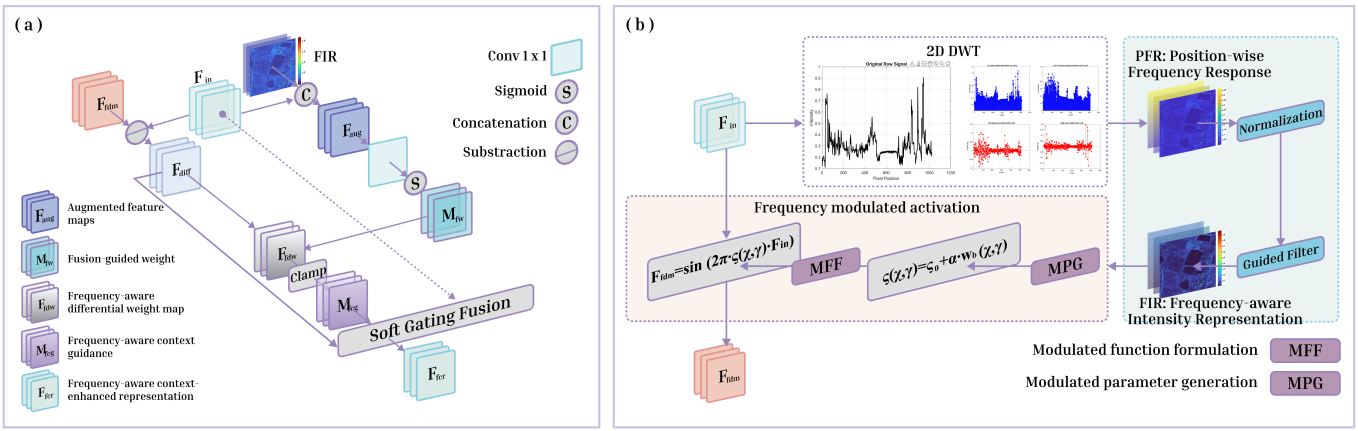

Fig. 2. Details of (a) FFM and (b) FDM.

III. Method

A. Overview of the Proposed FreDNet

As illustrated in Fig. 1, the proposed FreDNet adopts an encoder–decoder architecture tailored for semantic segmentation of RSIs. The design is motivated by the observation that HF components in RSIs are often associated with structural cues such as boundaries, edges, and fine textures, which play an important role in dense prediction tasks, especially for high-resolution overhead imagery. Motivated by this insight, FreDNet introduces a frequency-guided denoising-style refinement strategy that enhances both feature representation and cross-stage fusion through frequency-domain modulation.

In the encoding stage, the input image is processed by a series of downsampling convolutional layers, where each stage comprises a set of DRBs. Each DRB integrates a conventional spatial transformation branch with a frequency-guided refinement branch, aiming to recalibrate spatial semantics by selectively emphasizing structurally informative responses while suppressing noise-like or unstable activations. Compared to conventional convolutional blocks, the DRB structure facilitates richer structural encoding, particularly around object boundaries and spatial discontinuities.

The decoder symmetrically reconstructs the segmentation map using upsampling layers, where each decoding stage is also equipped with DRBs to maintain progressive semantic refinement. Instead of using direct skip connections, FreDNet employs FCFMs to bridge the encoder and decoder, introducing frequency-sensitive guidance to regulate multiscale feature alignment in a spatially adaptive manner. This design helps improve boundary localization and alleviates feature inconsistency across stages.

The output of the decoder is passed through a 1×1 convolution and softmax activation to yield the final dense prediction map. All frequency-aware components in FreDNet are implemented using the 2-D DWT, which decomposes features into multiscale subbands and provides an explicit yet lightweight mechanism for frequency-aware modulation.

Overall, FreDNet departs from purely spatial modeling by embedding frequency priors throughout the feature extraction and fusion pipeline. Rather than assuming a specific noise distribution, the proposed design performs frequency-guided feature refinement, which can be interpreted as a denoising-style process that promotes structural consistency and suppresses unstable responses.

B. Details of DRB

As illustrated in Fig. 1, the proposed DRB serves as a fundamental component of FreDNet for frequency-aware representation learning. Different from conventional residual blocks that focus solely on spatial transformations, DRB is designed to decouple spatial feature extraction and frequency-guided refinement within a unified residual framework. It consists of two parallel branches: a standard convolutional transformation branch and a frequency-guided denoising-style refinement branch composed of FDM and FFM, as illustrated in Fig. 2.

The final output of DRB is obtained via additive residual fusion

$$\mathbf{F}_{\text{drb}} = \mathbf{F}_{\text{in}} + \mathbf{F}_{\text{conv}} + \mathbf{F}_{\text{fcr}} \tag{1}$$

where $\mathbf{F}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$ denotes the input feature, $\mathbf{F}_{\text{conv}}$ represents the output of the standard convolutional block (top branch), and $\mathbf{F}_{\text{fcr}}$ denotes the frequency-aware context-refined representation produced by the bottom branch.

This residual formulation ensures stable optimization and compatibility with existing encoder–decoder backbones, while allowing frequency-guided refinement to be integrated without altering the overall network topology.

1) Frequency-Aware Denoising Module: The FDM is designed to encode HF spectral cues and to adaptively modulate spatial features for structural refinement. In this work, the term denoising refers to a frequency-guided refinement process that suppresses noise-like or unstable feature responses while preserving structurally meaningful HF information, rather than assuming an explicit noise model.

Given the input feature map $\mathbf{F}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, a 2-D DWT is applied independently along each channel to decompose the feature into frequency subbands

$$[\mathbf{F}_{\text{LL}}, \mathbf{F}_{\text{LH}}, \mathbf{F}_{\text{HL}}, \mathbf{F}_{\text{HH}}] = \text{DWT}(\mathbf{F}_{\text{in}}) \tag{2}$$

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where $\mathbf{F}_{\mathrm{LL}}$ denotes the low-frequency approximation, and $\mathbf{F}_{\mathrm{LH}}, \mathbf{F}_{\mathrm{HL}}, \mathbf{F}_{\mathrm{HH}} \in \mathbb{R}^{C\times(H/2)\times(W/2)}$ correspond to horizontal, vertical, and diagonal HF components, respectively.

Specifically, we adopt the Haar wavelet as the default basis for all DWT operations in FreDNet. This choice is motivated by its computational efficiency, orthogonality, and compact support, which make it particularly effective for capturing abrupt spatial transitions such as edges, contours, and object boundaries that frequently occur in high-resolution RSIs. In addition, the simple filter structure of the Haar wavelet facilitates stable integration into end-to-end trainable segmentation networks without introducing excessive computational overhead.

It is worth emphasizing that, in FreDNet, the DWT is not used as a handcrafted feature extractor, but rather as a structural frequency decomposition operator whose outputs are subsequently modulated by learnable components, including FDM, FFM, and FCFM. As a result, the network retains sufficient flexibility to adapt frequency responses during training, which alleviates sensitivity to a specific wavelet basis. While more complex wavelet bases (e.g., Daubechies wavelets) may capture smoother or higher order frequency patterns, the Haar wavelet provides an effective and efficient tradeoff between structural sensitivity and computational cost in the context of dense semantic segmentation. Exploring alternative wavelet bases is, therefore, considered a promising direction for future work.

To quantify local frequency intensity, we define a position-wise frequency response (PFR) map as

$$\mathbf{w}_b(x,y) = \sqrt{\mathbf{F}^2_{\mathrm{LH}}(x,y) + \mathbf{F}^2_{\mathrm{HL}}(x,y) + \mathbf{F}^2_{\mathrm{HH}}(x,y)} \tag{3}$$

which serves as a proxy for the aggregated HF energy at each spatial location.

Since raw frequency responses may contain outlier activations or unstable spectral fluctuations, we normalize $\mathbf{w}_b$ and apply a guided filter to obtain a smoothed yet structure-aware frequency intensity representation

$$\mathbf{F}_{\mathrm{FIR}} = \mathrm{GF}(\mathrm{Norm}(\mathbf{w}_b)) \tag{4}$$

where $\mathrm{GF}(\cdot)$ denotes the guided filtering using $\mathbf{F}_{\mathrm{in}}$ as the guidance image. In our implementation, the guided filter is configured with a radius of $r = 4$ and a regularization parameter $\epsilon = 10^{-3}$, following common practice in edge-preserving smoothing. The radius controls the local window size for filtering, while ϵ balances edge preservation and noise suppression. These parameters are fixed across all experiments and datasets. The resulting $\mathbf{F}_{\mathrm{FIR}} \in \mathbb{R}^{1\times(H/2)\times(W/2)}$ highlights spatial regions with relatively strong and structurally consistent frequency responses, which are subsequently used to guide frequency-aware modulation and fusion.

Instead of conventional activation functions, we introduce a frequency-modulated sinusoidal transformation

$$\mathbf{S}(x,y) = s_0 + \alpha \cdot \mathbf{w}_b(x,y) \tag{5}$$

$$\mathbf{F}_{\mathrm{fdm}} = \sin(2\pi \cdot \mathbf{S}(x,y) \cdot \mathbf{F}_{\mathrm{in}}) \tag{6}$$

where s_0 is a learnable base frequency initialized to 1.0, and α controls the sensitivity to local frequency intensity.

This sinusoidal modulation provides a smooth yet expressive nonlinear mapping conditioned on frequency intensity, enabling the network to adaptively emphasize or attenuate feature responses according to local spectral characteristics. The output $\mathbf{F}_{\mathrm{fdm}} \in \mathbb{R}^{C\times H\times W}$ constitutes the frequency-refined feature representation.

Overall, FDM transforms frequency priors into a spatially adaptive nonlinear modulation mechanism. By conditioning feature refinement on frequency intensity rather than hard thresholds or explicit noise assumptions, FDM offers a flexible and data-driven approach to structural enhancement in RSI segmentation.

2) Frequency-Aware Fusion Module: The FFM is designed to selectively integrate the frequency-refined features from FDM with the original spatial representation, further improving structural consistency while suppressing residual noise-like responses.

Given the input feature $\mathbf{F}_{\mathrm{in}} \in \mathbb{R}^{C\times H\times W}$ and the refined feature $\mathbf{F}_{\mathrm{fdm}} \in \mathbb{R}^{C\times H\times W}$, we first compute their residual discrepancy

$$\mathbf{F}_{\mathrm{diff}} = \mathrm{ReLU}(\mathbf{F}_{\mathrm{fdm}} - \mathbf{F}_{\mathrm{in}}) \tag{7}$$

which captures frequency-guided enhancement responses while discarding negative differences that may correspond to suppressed or unstable activations.

To guide fusion, we construct an augmented representation by concatenating $\mathbf{F}_{\mathrm{in}}$ with the frequency-aware intensity map $\mathbf{F}_{\mathrm{FIR}}$

$$\mathbf{F}_{\mathrm{aug}} = \mathrm{Concat}(\mathbf{F}_{\mathrm{in}}, \mathbf{F}_{\mathrm{FIR}}) \in \mathbb{R}^{(C+1)\times H\times W}. \tag{8}$$

A 1×1 convolution followed by a sigmoid activation is applied to learn a soft fusion weight map:

$$\mathbf{M}_{\mathrm{fw}} = \sigma\left(\mathrm{Conv}_{1\times 1}\left(\mathbf{F}_{\mathrm{aug}}\right)\right). \tag{9}$$

The weighted discrepancy is computed as

$$\mathbf{F}_{\mathrm{fdw}} = \mathbf{M}_{\mathrm{fw}} \odot \mathbf{F}_{\mathrm{diff}}. \tag{10}$$

To regulate the strength of frequency-guided enhancement and ensure stable optimization, we introduce a scaling factor β followed by a clamping operation:

$$\mathbf{M}_{\mathrm{fcg}} = \mathrm{Clamp}(\beta \cdot \mathbf{F}_{\mathrm{fdw}}, 0, 1) \tag{11}$$

where β is empirically set to 0.5.

Finally, the output of FFM is obtained via soft-gated residual fusion

$$\mathbf{F}_{\mathrm{fcr}} = \mathbf{F}_{\mathrm{in}} + \mathbf{M}_{\mathrm{fcg}} \odot \mathbf{F}_{\mathrm{diff}}. \tag{12}$$

This selective fusion strategy allows frequency-enhanced responses to be injected in a controlled manner, preserving spatial context while mitigating the risk of over-enhancement.

C. Details of FCFM

To enhance semantic consistency and structural alignment between encoder and decoder stages, we propose the FCFM, as illustrated in Fig. 1. Unlike conventional skip connections that directly propagate features, FCFM introduces frequency-sensitive guidance to regulate cross-stage fusion in a lightweight and spatially adaptive manner.

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TABLE I
EXPERIMENTAL SETTINGS

Items	Settings
Learning strategy	Adam
Initial learning rate	0.02
Loss function	Cross-entropy
Max epoch	500
GPU memory	48 GB
Batch size	32

Let $\mathbf{F}_{\text{enc}} \in \mathbb{R}^{C\times H\times W}$ and $\mathbf{F}_{\text{dec}} \in \mathbb{R}^{C\times H\times W}$ denote encoder and decoder features at the same resolution level. Frequency-aware guidance is derived from $\mathbf{F}_{\text{enc}}$ using a 2-D DWT

$$\mathbf{w}_b(x,y) = \sqrt{\mathbf{F}^2_{\text{LH}}(x,y) + \mathbf{F}^2_{\text{HL}}(x,y) + \mathbf{F}^2_{\text{HH}}(x,y)}. \tag{13}$$

After normalization and guided filtering, we obtain the frequency-aware intensity representation

$$\mathbf{F}_{\text{FIR}} = \text{GF}(\text{Norm}(\mathbf{w}_b)) \in \mathbb{R}^{1\times H\times W}. \tag{14}$$

This representation is transformed into a spatial attention map

$$\mathbf{M}_{\text{fir}} = \sigma\left(\text{Conv}_{1\times 1}(\mathbf{F}_{\text{FIR}})\right) \tag{15}$$

which softly highlights frequency-sensitive regions.

Encoder and decoder features are concatenated and modulated by the attention map

$$\mathbf{F}_{\text{cat}} = \text{Concat}(\mathbf{F}_{\text{enc}}, \mathbf{F}_{\text{dec}}) \tag{16}$$

$$\mathbf{F}_{\text{mod}} = \mathbf{F}_{\text{cat}} \odot \mathbf{M}_{\text{fir}}. \tag{17}$$

A 1×1 convolution projects the modulated features back to the original channel dimension

$$\mathbf{F}_{\text{fcfm}} = \text{Conv}_{1\times 1}(\mathbf{F}_{\text{mod}}). \tag{18}$$

By injecting frequency-sensitive guidance into cross-stage fusion, FCFM helps alleviate feature misalignment and improves the preservation of fine structural details during decoding, while maintaining a lightweight and architecture-agnostic design.

IV. EXPERIMENTS AND RESULTS

A. Settings

All experiments are conducted on a Linux server with an NVIDIA A40 GPU (48 GB). The proposed FreDNet is implemented using PyTorch, and trained end-to-end from scratch. The hyperparameters are empirically tuned for stability and performance. The optimization and training configuration are listed in Table I.

We reproduced 12 comparative methods, including DeepLab V3+ [21], DANet [31], RAANet [44], MACU-Net [45], SCAttNet [39], A2FPN [46], ST-UNet [56], FsaNet [60], RS3Mamba [3], BEDSN [75], SFFNet [93], and AFE-Net [97], covering CNN-based, transformer-based, and frequency-domain-incorporated methods.

B. Datasets

ISPRS Vaihingen: The ISPRS Vaihingen dataset [64] contains high-resolution aerial imagery captured over the city of Vaihingen, Germany. The data have a spatial resolution of 9 cm/pixel and includes 33 top–down true orthophotos, each approximately 2494×2064 pixels in size. The available spectral channels include RGB, and the semantic labels cover six land-cover classes: impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background. Following common practice, we omit the clutter category during evaluation. The high intraclass variation and fine boundary structures in this dataset make it well-suited for assessing detail-preserving segmentation models.

ISPRS Potsdam: The ISPRS Potsdam dataset [65] provides 38 aerial images with an even finer ground sampling distance (GSD) of 5 cm. Each image is of size 6000×6000 pixels and includes four spectral bands (near-infrared, R, G, and B), along with auxiliary data. In our experiments, only the RGB channels are utilized to ensure fair comparison with existing RGB-based methods. The label taxonomy mirrors that of Vaihingen, comprising six categories. The dataset poses considerable challenges due to its large image size, class imbalance, and subtle differences between categories.

LoveDA: The LoveDA dataset [66] is designed for dense urban–rural scene understanding and contains satellite images acquired via the Google Earth platform. The dataset spans over 536 km^2 across three cities in China: Nanjing, Changzhou, and Wuhan. Each image has a spatial resolution of 0.3 m and a size of 1024×1024 pixels. It includes seven semantic classes: background, building, road, water, barren, forest, and agriculture. The dataset is known for its high intraclass variability, large-scale coverage, and class imbalance, particularly in rural settings, making it a highly challenging benchmark.

To handle memory and computational constraints while ensuring diverse training samples, all images are cropped into nonoverlapping patches of size 256×256 pixels. We adopt a consistent dataset split across all three benchmarks using a 6:2:2 ratio for training, validation, and testing, respectively. This ensures statistically meaningful evaluation while maintaining representative scene distributions across splits.

C. Evaluation Metrics

To quantitatively assess segmentation performance, we adopt four widely used evaluation metrics: classwise F_1 score, average F_1 score (AF), overall accuracy (OA), and mean intersection over union (mIoU). These metrics jointly reflect the model's ability to distinguish categories, maintain overall consistency, and capture fine-grained boundaries.

The F_1 score is the harmonic mean of precision and recall, used to evaluate the segmentation quality per class

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{19}$$

Here, Precision = (TP/TP+FP) and Recall = (TP/TP+FN), where TP, FP, and FN denote the number of true positive, false positive, and false negative pixels, respectively.

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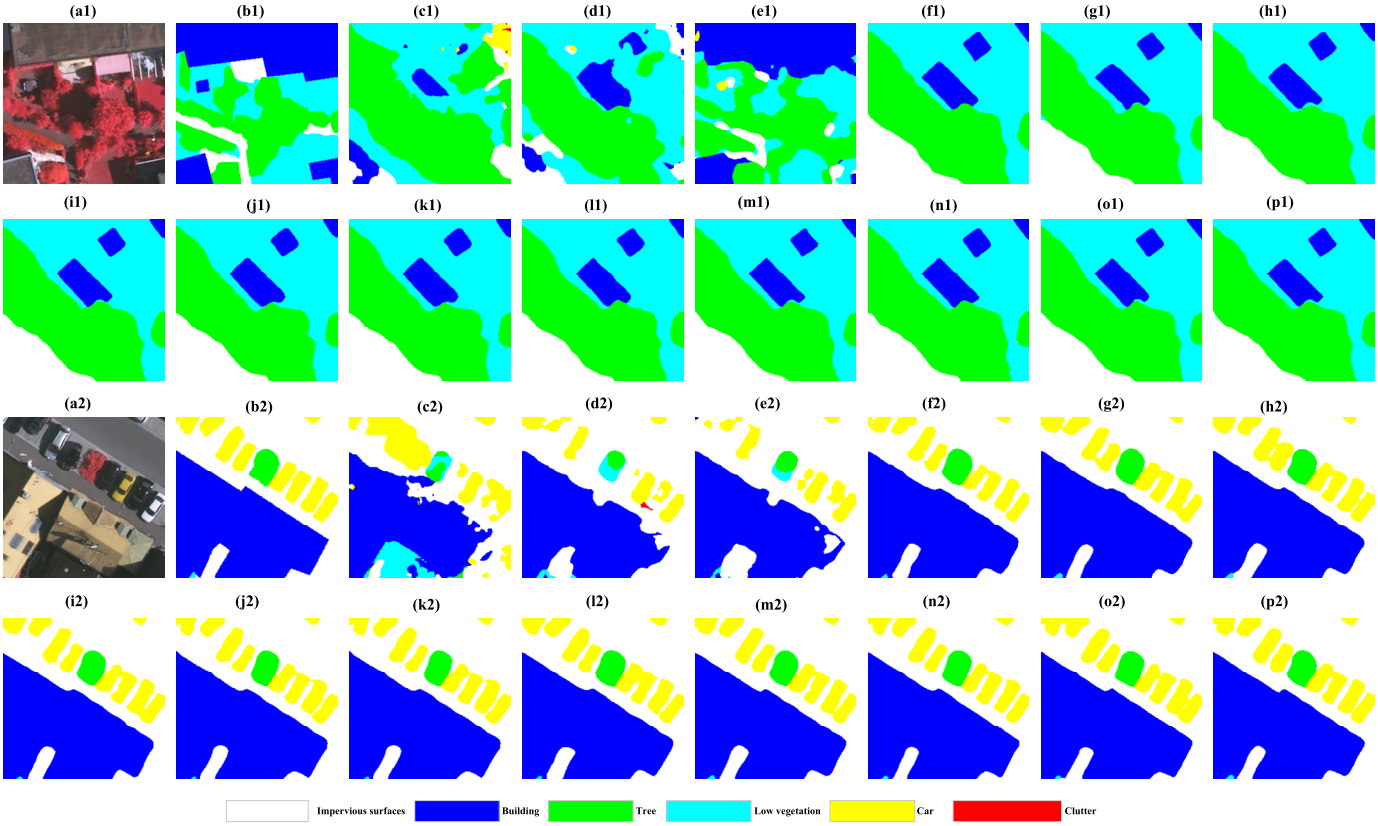

Fig. 3. Visualization comparisons on the ISPRS Vaihingen test set. (a) Input, (b) ground truth, (c) DeepLab V3+ [21], (d) DANet [31], (e) RAANet [44], (f) MACU-Net [45], (g) SCAttNet [39], (h) A2FPN [46], (i) ST-UNet [56], (j) FsaNet [60], (k) CLCFormer [54], [1] RS3Mamba [3], (m) BEDSN [75], (n) SFFNet [93], (o) AFE-Net [97], and (p) FreDNet (ours).

TABLE II

RESULTS ON THE ISPRS VAIHINGEN DATASET. CLASSWISE F_1 SCORE, AF, OA, AND MIOU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU
DeepLab V3+ [21]	87.99	87.80	72.72	85.55	47.37	76.29	74.47	69.69
DANet [31]	91.57	90.37	80.01	88.15	58.50	79.32	77.50	72.46
RAANet [44]	90.09	89.72	76.22	86.13	48.55	78.14	75.33	70.49
MACU-Net [45]	89.52	90.11	74.32	87.15	62.70	80.76	77.86	72.86
SCAttNet [39]	90.44	91.63	81.22	81.49	71.54	83.26	81.52	75.29
A2FPN [46]	92.99	95.17	80.98	89.25	87.62	89.20	87.07	81.06
ST-UNet [56]	91.52	92.72	82.19	82.46	72.39	84.26	82.29	76.01
FsaNet [60]	93.81	95.70	82.93	91.94	88.44	90.56	88.48	81.72
CLCFormer [54]	90.01	92.26	77.17	87.97	80.08	85.50	83.46	78.10
RS3Mamba [83]	93.18	94.92	85.16	89.37	91.89	90.90	89.12	82.78
BEDSN [75]	**94.71**	92.30	83.71	89.24	86.31	89.25	86.93	80.82
SFFNet [93]	93.02	94.15	81.63	88.44	84.66	88.78	87.10	80.31
AFE-Net [97]	93.37	94.86	84.21	89.01	87.73	89.84	88.65	81.73
FreDNet (ours)	93.66	**95.38**	**85.62**	**89.94**	**92.07**	**91.33**	**89.57**	**83.31**

The AF is computed by taking the mean of classwise F_1 scores across all semantic categories

$$\mathrm{AF} = \frac{1}{C}\sum_{i=1}^{C} F_1^{(i)} \tag{20}$$

where C is the number of classes.

The OA measures the proportion of correctly predicted pixels over the entire test set

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}. \tag{21}$$

Here, TN represents the number of true negative pixels.

The mIoU quantifies the overlap between the predicted and ground-truth regions for each class. The mIoU is calculated as

$$\mathrm{IoU}^{(i)} = \frac{\mathrm{TP}^{(i)}}{\mathrm{TP}^{(i)}+\mathrm{FP}^{(i)}+\mathrm{FN}^{(i)}}, \quad \mathrm{mIoU} = \frac{1}{C}\sum_{i=1}^{C}\mathrm{IoU}^{(i)}. \tag{22}$$

D. Comparative Experiments

1) Results on the ISPRS Vaihingen Dataset: We compare the proposed FreDNet against a comprehensive set of baselines on the ISPRS Vaihingen dataset. Quantitative results are summarized in Table II, while Fig. 3 provides qualitative comparisons.

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TABLE III
RESULTS ON THE ISPRS POTSDAM DATASET. CLASSWISE F_1-SCORE, AF, OA, AND MIOU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Impervious surfaces	Building	Low vegetation	Tree	Car	AF	OA	mIoU
DeepLab V3+ [21]	84.49	86.13	77.37	77.37	85.47	82.17	80.82	73.59
DANet [31]	86.96	92.04	80.14	79.95	89.36	85.69	83.99	77.13
RAANet [44]	90.40	95.70	87.20	81.80	77.60	86.54	85.21	77.95
MACU-Net [45]	89.16	92.76	86.80	83.02	79.25	86.20	85.90	77.99
SCAttNet [39]	91.87	96.90	85.24	87.05	92.78	90.77	89.06	80.83
A2FPN [46]	90.67	95.69	85.54	85.44	91.55	89.78	88.25	80.67
ST-UNet [56]	92.75	97.82	86.05	87.87	93.66	91.63	90.39	81.60
FsaNet [60]	93.18	98.28	86.46	88.29	94.10	92.06	90.61	81.79
CLCFormer [54]	92.66	96.64	87.05	88.00	93.69	91.61	89.97	81.68
RS3Mamba [83]	93.10	96.55	86.71	88.21	94.02	91.72	90.04	82.12
BEDSN [75]	**93.89**	**97.93**	86.33	88.09	94.21	92.09	90.71	82.31
SFFNet [93]	92.38	96.11	86.42	87.84	93.41	91.23	89.74	81.59
AFE-Net [97]	92.87	96.46	86.57	88.03	93.82	91.55	89.95	81.87
FreDNet (ours)	93.45	96.89	**87.14**	**88.55**	**94.33**	**92.07**	**90.53**	**82.76**

As shown in Table II, FreDNet achieves competitive performance across all evaluation metrics, including AF, OA, and mIoU. Specifically, it reports an AF score of 91.33, an OA of 89.57%, and an mIoU of 83.31%, showing consistent improvements over a range of recent methods. Compared with RS3Mamba, FreDNet yields gains of +0.43 in AF, +0.45% in OA, and +0.53% in mIoU, while also outperforming transformer-based approaches such as AFE-Net and BEDSN in overall segmentation accuracy. These results suggest that the proposed frequency-guided design provides an effective complement to spatial-domain modeling for complex high-resolution scenes.

A closer examination of the classwise F_1 scores reveals that FreDNet performs particularly well on categories characterized by fine structures or sharp boundaries. Specifically, FreDNet attains strong results on impervious surfaces (93.66), building (95.38), low vegetation (85.62), and car (92.07). For the tree category, FreDNet achieves a score of 89.94, which is comparable to that of RS3Mamba, indicating stable performance on vegetation-dominated regions. The balanced improvements across both large-area classes and small or thin objects indicate that FreDNet can maintain semantic consistency while preserving structural details.

These quantitative trends can be partially attributed to the integration of frequency-guided refinement and cross-stage modulation within DRB and FCFM. By introducing frequency-aware cues into both feature extraction and fusion, FreDNet facilitates the preservation of boundary-related information and helps alleviate feature misalignment across scales, which is particularly beneficial for high-resolution RSIs.

Fig. 3 provides qualitative comparisons that further support the quantitative observations. Compared to previous methods such as DeepLab V3+, DANet, and A2FPN, FreDNet produces segmentation maps with clearer object boundaries and more coherent region delineation. The improvements are especially noticeable for small objects such as cars and for narrow or elongated structures within impervious surface regions. In contrast, several competing methods exhibit fragmented predictions or boundary ambiguity in these challenging areas.

Overall, both numerical results and visual comparisons indicate that FreDNet achieves a robust balance between global semantic understanding and fine-grained structural preservation on the ISPRS Vaihingen dataset.

2) Results on the ISPRS Potsdam Dataset: Table III presents the performance comparison of FreDNet with several SOTA methods on the ISPRS Potsdam dataset. FreDNet demonstrates competitive performance across multiple evaluation metrics on this benchmark. In particular, it achieves an AF score of 92.07, which is comparable to those of recent high-performing methods such as RS3Mamba (91.72) and BEDSN (92.09). In terms of OA and mIoU, FreDNet attains 90.53% and 82.76%, respectively, indicating consistent improvements in pixel-level classification accuracy and region-level consistency.

On classwise evaluation, FreDNet yields leading or highly competitive F_1 scores across all five categories. For example, our model reports 93.45 on impervious surfaces, 96.89 on buildings, and 94.33 on cars, which are comparable to or slightly higher than those achieved by the best-performing counterparts. For vegetation-related categories such as trees and low vegetation, FreDNet achieves F_1 scores of 88.55 and 87.14, respectively, reflecting balanced performance across both fine-grained objects and larger terrain structures. Compared with transformer-based approaches such as CLCFormer and RS3Mamba, these results suggest that FreDNet maintains competitive global contextual modeling while providing improved sensitivity to boundary-related details.

Fig. 4 presents qualitative comparisons of segmentation results produced by different methods. As shown in Fig. 4(o), FreDNet generates segmentation maps with clearer boundaries and more distinct separation between adjacent categories, such as car versus building and tree versus low vegetation. In contrast, several baseline models, including DeepLab V3+, DANet, and RAANet, exhibit boundary leakage or category confusion in densely structured or cluttered regions. These visual differences indicate that the frequency-guided modules in FreDNet can help preserve structural integrity and mitigate noise-like artifacts across encoder–decoder stages.

Overall, the quantitative and qualitative results on the ISPRS Potsdam dataset indicate that FreDNet provides a robust and generalizable solution for high-resolution aerial scene segmentation, achieving competitive performance across diverse object categories and scene configurations.

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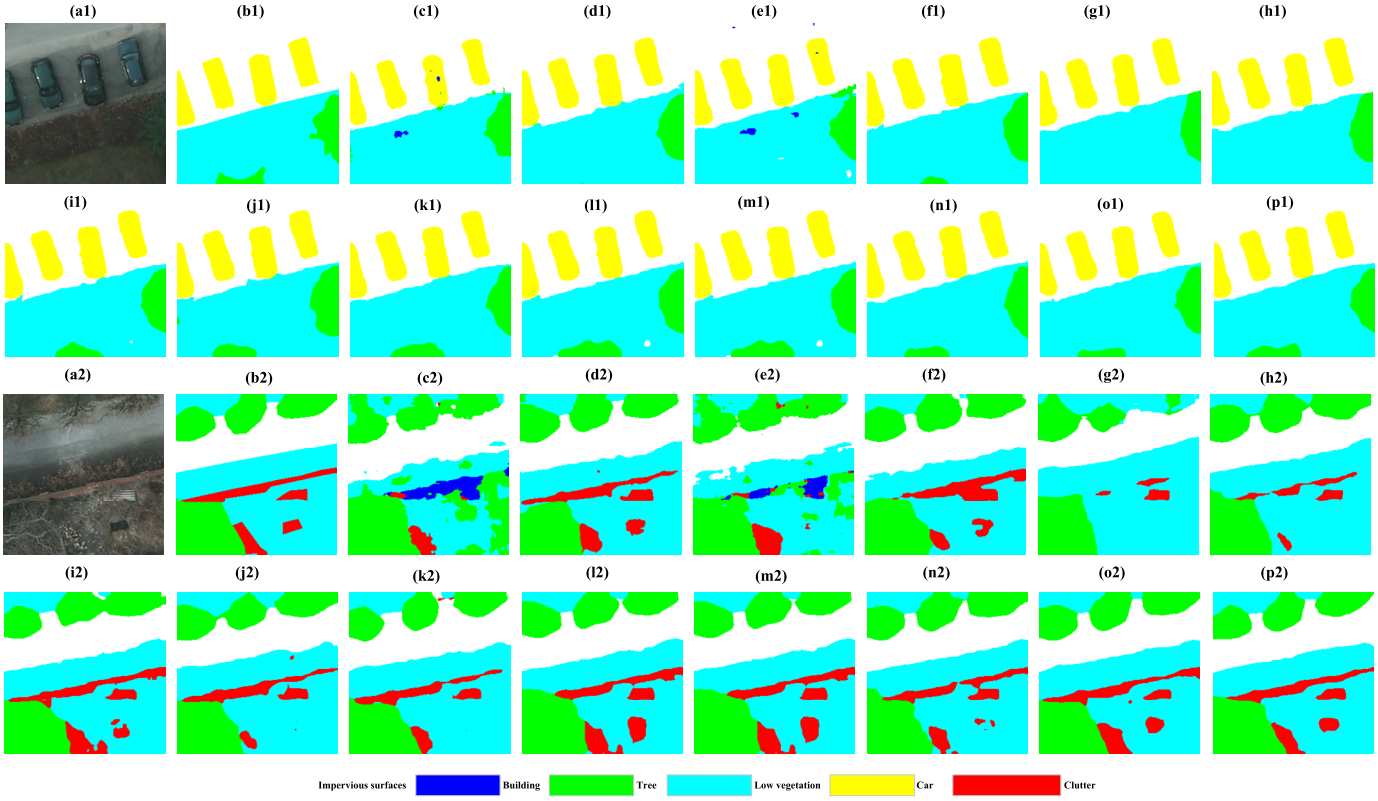

Fig. 4. Visualization comparisons on the ISPRS Potsdam test set. (a) Input, (b) ground truth, (c) DeepLab V3+ [21], (d) DANet [31], (e) RAANet [44], (f) MACU-Net [45], (g) SCAttNet [39], (h) A2FPN [46], (i) ST-UNet [56], (j) FsaNet [60], (k) RS3Mamba [3], (l) BEDSN [75], (m) SFFNet [93], (n) AFE-Net [97], and (o) FreDNet (ours).

TABLE IV

RESULTS ON THE LOVEDA DATASET. CLASSWISE F_1 SCORE, AF, OA, AND MIOU ARE LISTED, WHERE THE BOLD TEXT INDICATES THE BEST RESULTS

Methods	Background	Building	Road	Water	Barren	Forest	Agriculture	AF	OA	mIoU
DeepLab V3+ [21]	52.29	54.99	57.16	77.96	16.11	48.18	67.79	53.50	52.30	47.62
DANet [31]	54.47	61.02	63.37	79.17	26.63	52.28	70.02	58.14	54.64	50.18
RAANet [44]	55.02	62.19	65.58	81.03	29.25	54.11	74.07	60.18	58.95	53.93
MACU-Net [45]	59.16	64.08	66.73	81.01	32.23	55.81	75.79	62.12	59.65	54.16
SCAttNet [39]	65.95	71.88	77.04	86.61	50.79	61.19	82.00	70.78	67.31	61.09
A2FPN [46]	65.17	73.32	75.19	88.01	48.82	59.96	79.71	70.03	66.89	61.14
ST-UNet [56]	66.68	71.07	75.44	88.68	51.36	64.20	81.42	71.26	69.78	63.04
FsaNet [60]	68.63	75.95	79.38	89.61	53.47	66.32	82.71	73.72	70.59	63.49
CLCFormer [54]	67.17	74.34	77.69	87.71	52.34	64.91	80.96	72.16	70.45	62.55
RS3Mamba [83]	70.25	76.52	81.45	89.94	54.82	**70.68**	84.32	75.57	72.01	65.81
BEDSN [75]	72.33	80.18	**84.37**	**92.14**	56.62	69.02	86.47	77.02	73.39	66.71
SFFNet [93]	69.01	74.86	79.93	89.32	52.93	67.14	83.09	73.04	71.36	64.12
AFE-Net [97]	69.78	75.67	80.25	89.61	53.84	68.01	83.87	73.72	71.58	64.38
FreDNet (ours)	**73.50**	**80.64**	84.12	91.85	**57.02**	69.98	**86.51**	**77.66**	**73.89**	**67.12**

3) Results on the LoveDA Dataset: Table IV presents the quantitative performance comparisons on the challenging LoveDA dataset, which involves diverse land covers, complex backgrounds, and significant intraclass variation. FreDNet demonstrates competitive performance across all evaluation metrics on this dataset. Specifically, it achieves an average AF of 77.66, an OA of 73.89%, and an mIoU of 67.12%, indicating consistent improvements over a range of recent methods.

Compared with representative high-performing baselines, such as RS3Mamba and BEDSN, FreDNet shows gains of +2.09 and +0.64 in AF, respectively. In terms of mIoU, FreDNet improves upon BEDSN by +0.41 and exceeds RS3Mamba by +1.31, suggesting that the proposed frequency-guided design is beneficial for modeling diverse semantic structures under real-world imaging conditions.

An examination of the classwise F_1 scores reveals that FreDNet achieves strong performance on most semantic categories. Specifically, it attains high F_1 scores on Background (73.50), Building (80.64), Road (84.12), Water (91.85), Barren (57.02), and Agriculture (86.51). Notably, these categories include underrepresented or spectrally ambiguous classes such as background and barren land, which are commonly challenging for existing approaches. For the forest category, FreDNet

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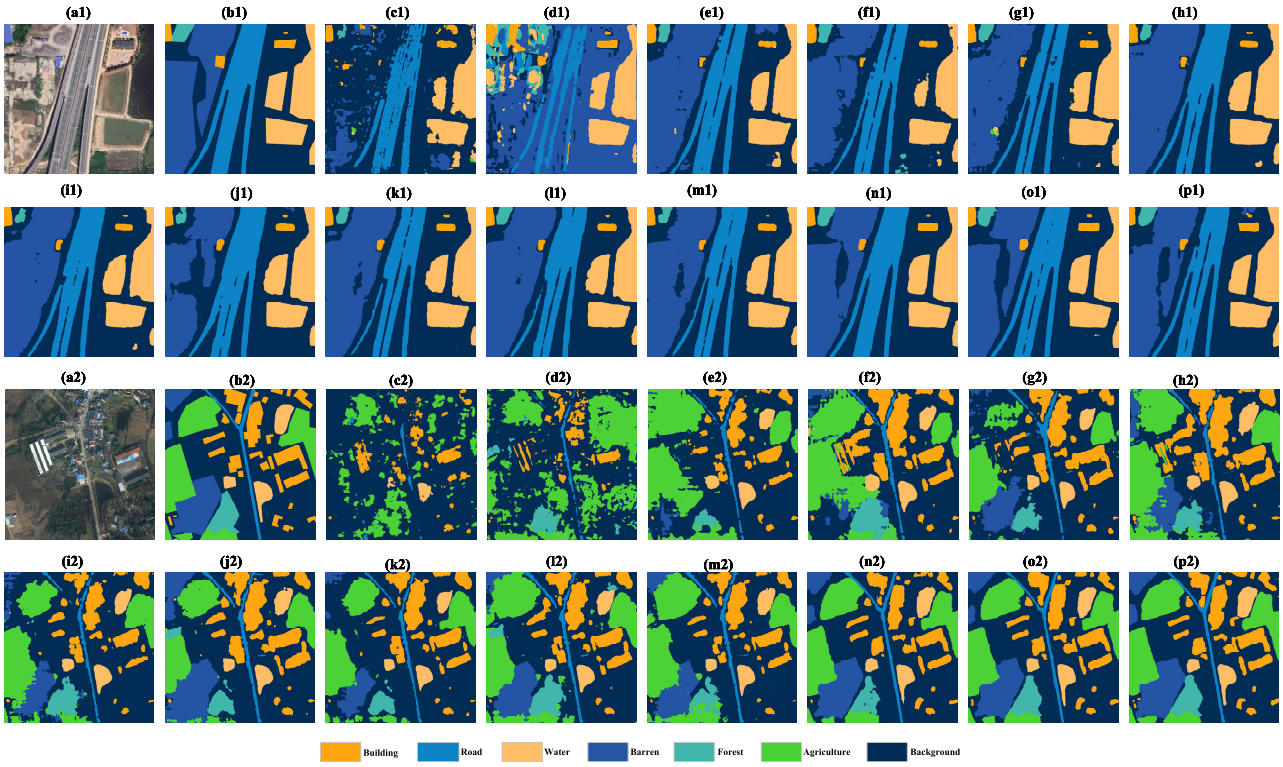

Fig. 5. Visualization comparisons on the LoveDA test set. (a) Input, (b) ground truth, (c) DeepLab V3+ [21], (d) DANet [31], (e) RAANet [44], (f) MACU-Net [45], (g) SCAttNet [39], (h) A2FPN [46], (i) ST-UNet [56], (j) FsaNet [60], (k) RS3Mamba [3], (l) BEDSN [75], (m) SFFNet [93], (n) AFE-Net [97], and (o) FreDNet (ours).

TABLE V

ABLATION STUDY OF DRB ON ISPRS VAIHINGEN, POTSDAM, AND LOVEDA DATASETS. RESULTS ARE REPORTED AS AF/OA(%)/MIOU(%)

Model	DRB Submodules		Datasets		
	FDM	FFM	Vaihingen	Potsdam	LoveDA
FreDNet	✓	✓	**91.33 / 89.57 / 83.31**	**92.07 / 90.53 / 82.76**	**77.66 / 73.89 / 67.12**
	✓		90.58 / 88.76 / 81.66	91.22 / 89.57 / 81.49	76.58 / 72.81 / 66.01
		✓	89.94 / 87.83 / 80.65	90.32 / 88.66 / 80.34	75.41 / 71.93 / 64.98
			88.26 / 86.32 / 78.24	89.01 / 87.04 / 79.27	73.62 / 70.34 / 63.15

achieves an F_1 score of 69.98, which is comparable to that of strong transformer-based methods, indicating balanced performance across both natural and man-made scenes.

Fig. 5 presents qualitative comparisons that further illustrate the behavior of different methods on complex LoveDA scenes. As shown in Fig. 5(o), FreDNet produces segmentation maps with relatively clearer semantic boundaries and improved preservation of fine-scale structures, such as road edges and building contours. In contrast, several baseline methods (e.g., DeepLab V3+, DANet, and MACU-Net) exhibit boundary ambiguity, semantic bleed-over, or overly smooth transitions, particularly in rural and mixed land-cover regions.

Overall, the quantitative and qualitative results on the LoveDA dataset indicate that FreDNet maintains robust performance under challenging distribution shifts and complex scene compositions, providing a favorable balance between global semantic understanding and fine-grained structural representation.

E. Ablation Study

1) Effects of DRB: To investigate the contribution of the proposed DRB, we conduct controlled ablation experiments by selectively enabling its two core submodules, namely, the FDM and the FFM. This design allows us to disentangle the individual and joint effects of frequency-guided refinement and fusion within the DRB framework. The quantitative results on the ISPRS Vaihingen, Potsdam, and LoveDA datasets are summarized in Table V.

When both FDM and FFM are disabled, the DRB degenerates into a standard residual block, resulting in the lowest performance across all datasets. Introducing either FDM or FFM individually leads to consistent performance improvements, indicating that each submodule contributes positively to representation refinement. Specifically, enabling FDM yields more noticeable gains than enabling FFM alone, suggesting that frequency-guided denoising-style refinement plays an important role in stabilizing feature responses and enhancing structurally informative cues. When both FDM and FFM are jointly activated, the model achieves the best overall performance, demonstrating that frequency-guided refinement and soft fusion are complementary within the DRB design.

To further analyze the qualitative impact of DRB and its submodules, Fig. 6 illustrates segmentation results and corresponding activation maps under four configurations: full

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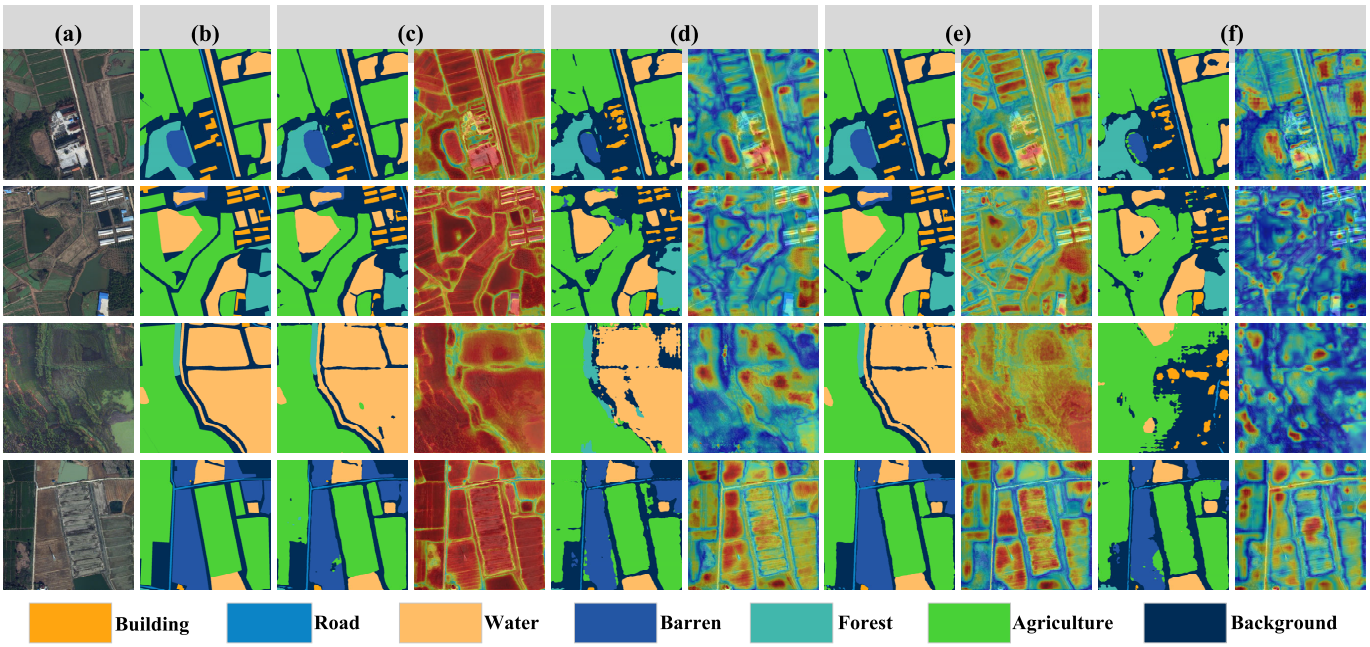

Fig. 6. Ablation study of submodules of DRB. (a) Input, (b) ground truth, (c) predicted by FreDNet (with FDM and FFM) and corresponding feature map of segmentation head, (d) predicted by FreDNet (without FDM) and corresponding feature map of segmentation head, (e) predicted by FreDNet (without FFM) and corresponding feature map of segmentation head, and (f) predicted by FreDNet (without FFM and FDM) and corresponding feature map of segmentation head.

TABLE VI
ABLATION RESULTS OF FCFM ON THREE DATASETS. PERFORMANCE IS REPORTED AS AF/OA(%)/MIOU(%)

Model	Fusion Strategy	Vaihingen	Potsdam	LoveDA
FreDNet	Element-wise Sum	89.72/87.20/80.91	90.03/88.37/81.12	75.94/72.08/65.94
	Concatenation	89.25/86.81/80.43	89.62/87.96/80.65	75.31/71.49/65.26
	FCFM (ours)	**91.33/89.57/83.31**	**92.07/90.53/82.76**	**77.66/73.89/67.12**

DRB, without FDM, without FFM, and without both modules. The full DRB configuration [Fig. 6(c)] produces clearer object boundaries and more coherent semantic regions, with activation responses concentrated on structurally meaningful areas. Removing FDM [Fig. 6(d)] leads to noisier boundary predictions, particularly in vegetation- and water-related regions, indicating that frequency-guided denoising-style refinement helps suppress unstable spectral responses. When FFM is removed [Fig. 6(e)], semantic fragmentation and region-level inconsistency become more evident, suggesting that effective fusion is necessary for maintaining cross-feature coherence. In the degenerated case without both modules [Fig. 6(f)], both prediction quality and activation patterns degrade substantially, highlighting the complementary roles of FDM and FFM in DRB.

2) Effects of FCFM: To examine the effect of the proposed FCFM, we replace FCFM with two commonly used fusion strategies: elementwise summation and feature concatenation. All other components of FreDNet are kept unchanged to ensure a fair comparison that isolates the contribution of cross-stage fusion. The quantitative results on the Vaihingen, Potsdam, and LoveDA datasets are reported in Table VI.

As shown in Table VI, replacing FCFM with either elementwise summation or concatenation leads to a consistent performance drop across all datasets. Although these conventional fusion strategies achieve reasonable results, they lack explicit frequency-aware guidance and spatial adaptivity, which may limit their ability to align encoder and decoder features effectively. In contrast, FCFM consistently yields better performance, suggesting that frequency-guided cross-stage modulation contributes to improved semantic alignment and structural preservation during decoding.

3) Analysis of Frequency-Aware Denoising: While the quantitative ablation results demonstrate the effectiveness of the proposed DRB and its submodules, they do not directly reveal how frequency-aware denoising influences prediction behavior in the spectral domain. To provide further insight into this aspect, we conduct an additional analysis that explicitly examines HF error characteristics under different DRB configurations.

Specifically, we analyze the HF error energy ratio on the LoveDA test set. Given a prediction error map (prediction minus ground truth), we compute its Fourier spectrum and measure the proportion of spectral energy located in the HF band. This ratio reflects the extent to which prediction errors are dominated by HF components, which are often associated with noisy boundaries, fragmented contours, or unstable fine-scale responses. All results are averaged over five independent runs to reduce randomness.

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TABLE VII

MODEL COMPLEXITY UNDER DIFFERENT DRB CONFIGURATIONS. RESULTS ARE REPORTED IN TERMS OF PARAMS (M), FLOPS (G), AND INFERENCE TIME PER 256 × 256 IMAGE (MS)

Model	DRB Submodules		Params (M)	FLOPs (G)	Inference Time (ms)
	FDM	FFM			
FreDNet	✓	✓	39.84	181.6	24.8
	✓		36.23	162.4	23.6
		✓	37.41	170.9	24.1
			33.58	151.2	22.4

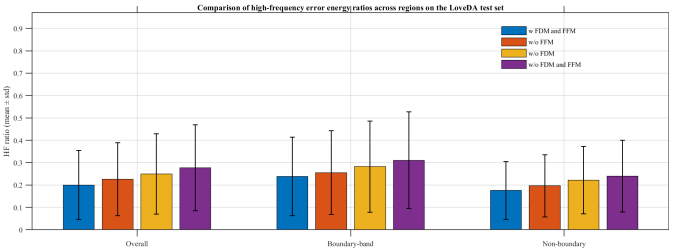

Fig. 7. Comparison of HF error energy ratios across different spatial regions on the LoveDA test set. Results are reported as mean ± standard deviation over five independent runs.

Fig. 7 shows the comparison of the HF error energy ratios across different spatial regions, including overall regions, boundary-band regions, and nonboundary regions, under four configurations: full DRB (with FDM and FFM), without FFM, without FDM, and without both modules. Across all region types, the full DRB configuration consistently exhibits the lowest HF error ratios, indicating that prediction errors contain relatively less HF energy. This trend is particularly evident in boundary-band regions, where removing FDM leads to a noticeable increase in HF error energy. This observation suggests that frequency-aware denoising-style refinement helps stabilize boundary predictions by suppressing spurious HF error components, rather than merely smoothing spatial outputs.

When FFM is removed, the HF error ratio also increases, although to a lesser extent than removing FDM, which aligns with the quantitative ablation results in Table V. In the degenerated configuration without both modules, the HF error energy is highest across all regions, indicating that the absence of frequency-aware refinement and fusion leads to more spectrally unstable prediction errors.

Overall, this analysis provides complementary spectral-domain evidence that the proposed frequency-aware denoising mechanism contributes to reducing HF error components, especially near object boundaries. Rather than enforcing excessive smoothing, the denoising behavior helps regulate unstable spectral responses, which is consistent with the improved boundary coherence and semantic consistency observed in both quantitative and qualitative results.

F. Complexity Analysis

To further assess the computational characteristics of different DRB configurations, we analyze the model complexity in terms of the number of parameters (Params), floating-point operations (FLOPs), and the average inference time per 256 × 256 input image. This analysis aims to quantify the additional computational overhead introduced by the frequency-aware components, rather than to claim absolute efficiency advantages. Table VII reports the results for FreDNet under four settings by selectively enabling or disabling the proposed FDM and FFM modules.

From Table VII, it can be observed that enabling the proposed FDM and FFM modules introduces a moderate increase in both parameter count and computational cost, while maintaining a comparable inference latency. Specifically, the full FreDNet configuration reaches 39.84M parameters and 181.6G FLOPs, with an average inference time of 24.8 ms/image under the same experimental setting. Compared with the configuration without DRB submodules, this increase reflects the additional frequency-aware processing rather than redundant architectural expansion.

When analyzed individually, FDM and FFM contribute to the overall complexity in different ways. FDM primarily introduces additional learnable parameters due to frequency-modulated activations and spectral-guided transformations, whereas FFM incurs relatively higher FLOPs due to its attention-guided feature fusion operations. Despite these differences, both submodules introduce incremental overhead that remains controlled within the overall encoder–decoder framework.

Combined with the performance gains reported in Table V, these results suggest that the proposed DRB design offers a favorable tradeoff between segmentation accuracy and computational cost. In particular, the added complexity is proportional to the improvements in boundary preservation and semantic consistency, supporting the practical feasibility of integrating frequency-aware refinement into RSI segmentation networks.

V. DISCUSSION AND CONCLUSION

In this article, we presented FreDNet, a frequency-guided denoising-style framework designed to enhance the robustness of HF structure modeling and cross-stage feature fusion in RSI semantic segmentation. Different from purely spatial-domain designs, FreDNet integrates spectral priors into both representation refinement and encoder–decoder fusion, aiming to improve boundary coherence and semantic consistency under complex scene composition. Through the proposed DRB, which integrates FDM and FFM pathways, the network regulates spectrally unstable responses and selectively preserves structurally informative details. Moreover, the FCFM module

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leverages frequency intensity response maps to promote cross-level semantic alignment, which is beneficial for fine-grained boundary delineation and contextual integration.

A. Discussion

A key motivation behind FreDNet is that segmentation errors in high-resolution RSIs are often coupled with unstable HF responses (e.g., fragmented contours and boundary jitter) and feature misalignment across encoder–decoder stages. In this context, the term "denoising" in FreDNet is not intended to imply a generic low-pass filtering of features. Instead, it describes a frequency-aware refinement process that aims to suppress spurious HF artifacts while retaining structurally meaningful details, guided by wavelet-derived intensity cues and learnable modulation. To make this behavior more transparent, we further provided ablation and analysis results, including an error-spectrum-based evaluation that measures the HF energy ratio of prediction error maps. This analysis offers complementary evidence that the proposed frequency-aware refinement can reduce HF error components, particularly in boundary-sensitive regions, without relying on indiscriminate smoothing. In addition, we reported complexity statistics and runtime measurements to clarify the practical overhead introduced by the frequency-aware modules. The results suggest that the gains brought by DRB and FCFM are achieved with a moderate and controllable computational cost, indicating that frequency-guided modeling can be incorporated without excessive architectural expansion.

The experimental results on three diverse benchmarks demonstrate that our method achieves strong and consistent performance compared to recent CNN- and transformer-based models in terms of F_1 score, OA, and mIoU. Across datasets, the frequency-aware components tend to be particularly beneficial for boundary-related structures and small-scale categories, where HF semantics and cross-stage alignment are more critical. These results support the view that frequency-sensitive design can serve as a complementary dimension to spatial-domain modeling in remote sensing segmentation.

B. Limitations and Future Work

Despite the encouraging results, several aspects remain open for further investigation. First, the current implementation adopts a fixed wavelet basis and a predefined frequency decomposition strategy; exploring adaptive or multilevel frequency basis learning may further improve flexibility across scene types and sensors. Second, while FreDNet is evaluated on RGB RSIs, extending the framework to multimodal settings (e.g., hyperspectral and LiDAR-assisted segmentation) may require modality-specific frequency modeling and fusion strategies. Finally, understanding how frequency-guided refinement interacts with large-scale pretraining or foundation-model paradigms is an interesting direction for future research.

In conclusion, FreDNet provides a frequency-guided segmentation framework that emphasizes spectral stability and cross-stage consistency as complements to conventional spatial-domain modeling. By integrating frequency-aware refinement and fusion mechanisms throughout the encoder–decoder pipeline, it offers a practical approach for improving fine-grained RSI segmentation, and we hope it encourages further exploration of frequency-aware representations in this field.

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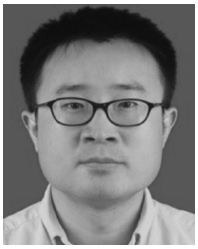

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